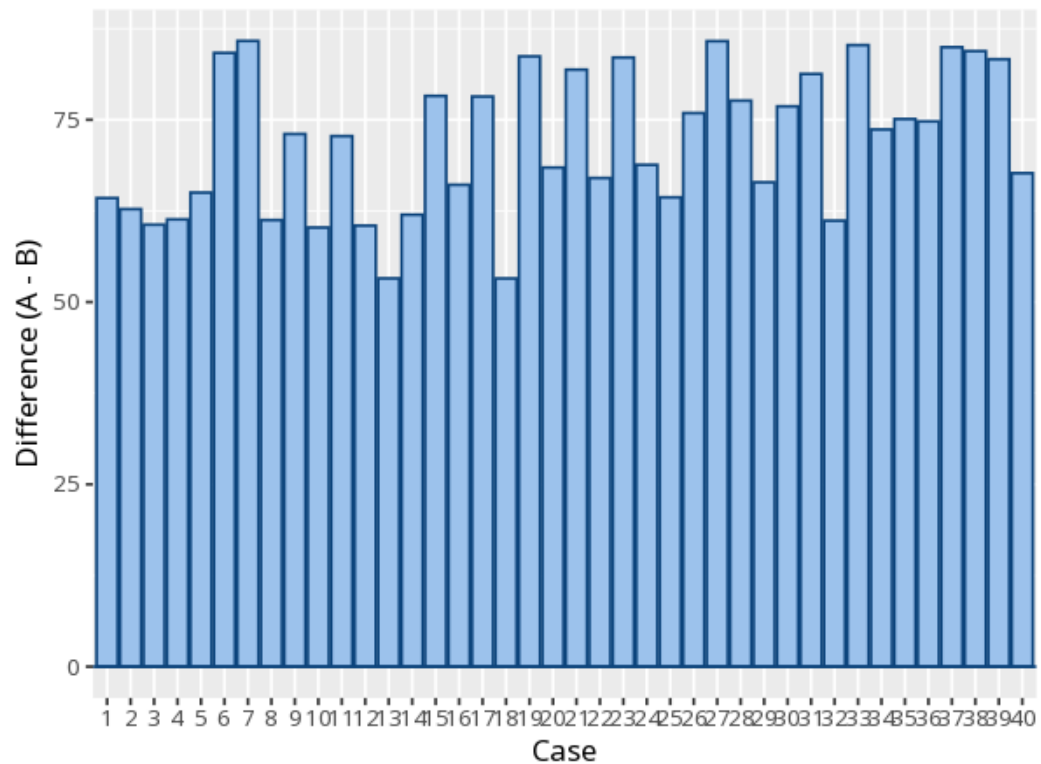


1 Wilcoxon signed-rank

Sign	N	Mean rank	Sum of ranks
Positive	40	20.50	820.00
Negative	0	—	—
Ties	0	—	—

V / W	Z	p	r [95% CI]	Median diff [95% CI]
820.00	5.51	<.001	1.00 [1.00, 1.00]	72.24 [68.63, 75.10]



The nonparametric counterpart to the paired t-test. It ranks the size of each paired change; a p below alpha means a systematic shift between the two conditions, with r as the effect size. The shift maps to the median difference only when the within-pair differences are roughly symmetric; the sign of the median difference (or Hodges–Lehmann estimate) gives the direction. Your significance threshold (α) is 0.05. Method: Wilcoxon signed rank exact test.

A Wilcoxon signed-rank test gave $V=820.00$, $Z=5.51$, $p < .001$, $r=1.00$ [1.00, 1.00].

Statistical basis: Wilcoxon signed-rank test (Wilcoxon, F. (1945). "Individual comparisons by ranking methods." Biometrics Bulletin, 1(6), 80-83.); Matched-pairs rank-biserial effect size (Ben-Shachar MS, Lüdtke D, Makowski

D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.).